Hierarchical Context Network for Airborne Image Segmentation

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***Abstract*— Most of the recent methods focus on capturing contextual information by measuring relations (e.g., feature similarity) between each pixel and all the others for airborne image segmentation. Nevertheless, these methods have difficulty in handling confusing objects with a partially similar appearance. In this article, we attempt to simultaneously explore pixel-to-pixel (P2P) and pixel-to-object (P2O) relations to learn contextual information. For this purpose, a hierarchical context network (HCNet) is proposed. It consists of a P2P subnetwork and a P2O subnetwork. The P2P subnetwork learns the P2P relation (detail-grained context) for better preservation of the details (e.g., boundary) of the objects. Meanwhile, the P2O subnetwork models the P2O relation (semantic-grained context), aiming at improving the intraobject semantic consistency. When inferring the segmentation results, outputs of these two subnetworks are aggregated to obtain the hierarchical contextual information. Experimental results demonstrate that the proposed model achieves competitive performance on three challenging benchmarks.**

***Index Terms*— Airborne images, hierarchical context, pixel-to-object (P2O) subnetwork, pixel-to-pixel (P2P) subnetwork, semantic segmentation.**

I. Introduction

SEGMENTATION with high-resolution airborne images is fundamental and significant to various applications, such as urban change detection [1], [2], disaster relief [3], and precision agriculture [4]. The goal of this task is to infer the semantic category of each pixel in the image. In the high-resolution airborne image, huge quantity of detailed information is contained, and objects show large variations in the scale and appearance, making image segmentation a very challenging task. Exploring contextual information [5]–[8] has been well recognized as an effective way to deal with this task.

In the past few years, a convolutional neural network (CNN) [9] has been the primary choice to capture contextual information. Early methods (e.g., FCN-8s [10], UNet [11], and SegNet [12]) based on CNN attempt to learn contextual information via an encoder–decoder structure. Although these methods succeed in using convolutional kernels to capture contextual information, their performances are still limited by the small receptive field of convolutional kernels [13]. Numerous works attempt to fuse contexts captured by convolutional kernels of different-scale receptive fields. Zhao *et al.* [14] proposed a pyramid pooling module to capture contextual information of different scales using several pooling kernels with different strides. Similarly, atrous spatial pyramid pooling, which consists of parallel dilated convolutions with different dilated rates, are utilized to encode multiscale contextual information [15]. Lin *et al.* [16] aggregated pairs of features between successive scales via bidirectional long short-term memory (LSTM) [17] for multiscale context fusion. Ding *et al.* [18] proposed a context contrasted local module to extract and integrate multilevel features and then proposed a scheme of the gated sum to selectively aggregate different scale of features from all layers. Fu *et al.* [19] stacked multiple shallow deconvolutional networks units with dense connection to integrate multiscale features.

In addition, several works propose to directly exploit relations (e.g., feature similarity) between the pixel and its contextual pixels regardless of their distances. For example, Zhang *et al.* [20] proposed to learn the contextual information via measuring the similarity between the target pixel and all the others, indicating how likely they would co-occur.

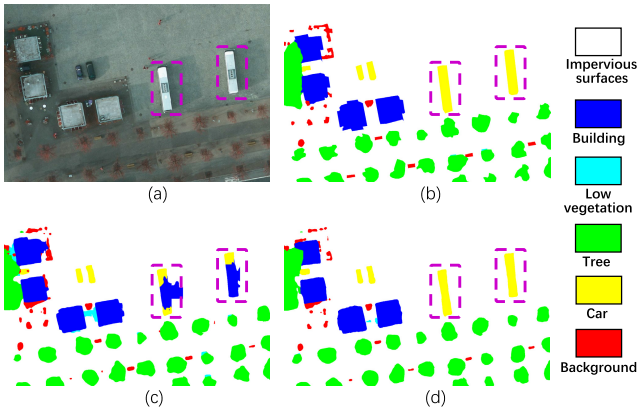


Fig. 1. (a) Airborne image (input image) where the marked *car* look similar to *building* and *road*. (b) Ground truth of (a). (c) Segmentation result of DANet. (d) Segmentation result of the proposed HCNet.

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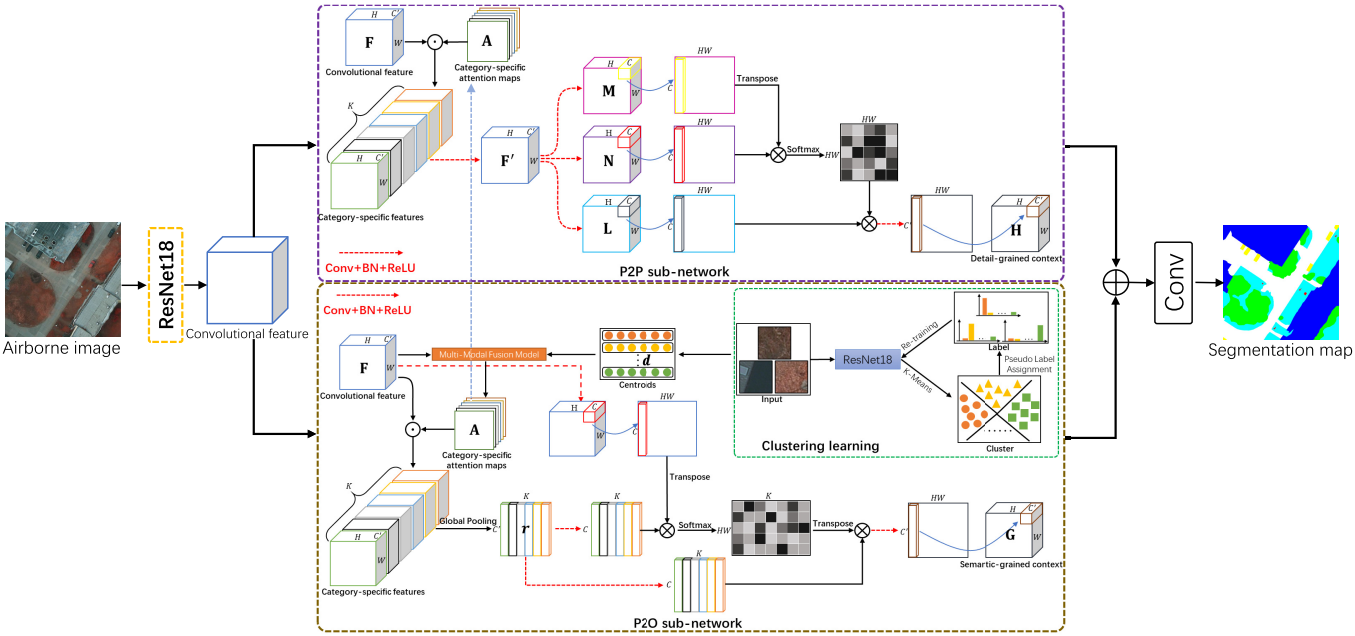


Fig. 2. Overall framework of HCNet for airborne image segmentation. H: the height of the feature map; W: the width of the feature map; C': the channel number of the convolutional feature; C: the channel number of the intermediate feature; and K: the number of categories.

Fu *et al.* [21] obtained the contextual information of each pixel by aggregating its contextual pixels via self-attention mechanism. Zhao *et al.* [22] connected each pixel of the feature map to all the other ones through a self-adaptively learned attention mask for obtaining long-range contextual information. Zhang *et al.* [23] proposed to learn the semantic context of scenes based on the relations between semantic representations of pixels. In order to capture nonlocal contextual information, a relation-augmented FCN, which models the short- and long-range relations among all pixels of the feature map, was proposed in [24]. However, all these methods attempt to distinguish different-category objects with partially similar appearances through the pixel-to-pixel (P2P) relation. That is, pixels lying in the different-category object regions may share similar feature representations, leading to imprecise segmentation of objects produced by P2P relation. As shown in Fig. 1(a), the appearance of the marked *car* is partially similar to those of the *building* and the surrounding *road*, so it is easily misclassified into the *building* and *road*, when only P2P relation is exploited in DANet [21].

In this article, we design two subnetworks (i.e., P2P subnetwork and pixel-to-object (P2O) subnetwork) to handle this issue. These two subnetworks are appended to ResNet18 [25], constituting hierarchical context network (HCNet) (as shown in Fig. 2). The goal of the P2P subnetwork is to capture detail-grained contextual information, which is utilized to describe the spatial details of the object. Meanwhile, the P2O subnetwork aims at extracting semantic-grained contextual information, which is responsible for providing semantic information for classifying the object. By aggregating detail-grained and semantic-grained contextual information, we could obtain hierarchical contextual information, which is discriminative enough to help improve the performance of the distinction of confusing objects. Fig. 1(d) has demonstrated that our model can correct the regions of the marked *car* and obtain sharp, unambiguous boundaries between the *car* and the *road*. Extensive ablation studies and comparison experiments with state-of-the-art benchmark models and related deep methods on two ISPRS datasets and Gaofen Image Dataset (GID) [26] validate that HCNet is effective.

II. Hierarchical Context Network

An overview of HCNet is given in Fig. 2. HCNet is mainly composed of ResNet18, P2P subnetwork, and P2O subnetwork. ResNet18 is served as a backbone network to extract the convolutional feature. Herein, we exclude the fully connected layer from ResNet18 to preserve spatial information. The convolutional feature is subsequently fed into the P2P subnetwork and the P2O subnetwork. In the P2P subnetwork, the detail-grained contextual information is extracted by using P2P relation. Meanwhile, P2O relation is modeled in the P2O subnetwork for constructing semantic-grained contextual information. These two kinds of contextual information are aggregated to form hierarchical contextual information. Finally, the hierarchical contextual information is fed into a 1×1 convolutional layer to obtain the segmentation map. To capture multiscale contextual information, the outputs of the last three layers of ResNet18 are, respectively, utilized to derive the corresponding segmentation map, and these three segmentation maps are fused in the form of weighted summation.

A. P2P Subnetwork

Recently, several works have used the self-attention mechanism to exploit P2P relation and attained a measure of success [20]–[24]. In these methods, the feature at each position is updated via aggregating features at all positions according to the corresponding feature similarities. Consequently, the features similar to the target feature would be highlighted

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Fig. 3. Examples of input images for clustering learning.

in the updated representation. However, two adjacent objects (e.g., car and road shown in Fig. 1) in different categories could easily cause ambiguous boundaries between them when they exhibit similar visual appearance. This is due to the spurious correlation between their corresponding features.

To alleviate this issue, we design a P2P subnetwork, as shown in Fig. 2. It aims at suppressing the above spurious correlation. Given the category-specific attention map $\mathbf{A}_k$ (the procedure of obtaining it will be given in the next subsection), we multiply it by each channel of the convolutional feature $\mathbf{F}$, emphasizing the feature of the kth category. After that, we aggregate multiple category-specific features to get a global category-level representation $\mathbf{F}'$ for each pixel via a convolutional layer followed by batch normalization and ReLU. Due to integrating category-level information into each pixel representation, $\mathbf{F}'$ could to some extent eliminate the negative effects of the visual similarity. Given the feature $\mathbf{F}'$, self-attention is used to derive the P2P relation. Specifically, we feed $\mathbf{F}'$ into two functions η and θ to separately generate two new features: $\mathbf{M} \in \mathbb{R}^{C\times H\times W}$ and $\mathbf{N} \in \mathbb{R}^{C\times H\times W}$. In addition, we reshape them to $\mathbf{M} \in \mathbb{R}^{C\times S}$ and $\mathbf{N} \in \mathbb{R}^{C\times S}$, where $S = H \times W$. Then, the transposes of $\mathbf{M}$ and $\mathbf{N}$ are combined via matrix multiplication, and softmax is performed on the result to obtain P2P relation $\mathbf{W}' \in \mathbb{R}^{S\times S}$

$$w'_{ij} = \frac{e^{\text{sim}(m_i, n_j)}}{\sum_{j=1}^{K} e^{\text{sim}(m_i, n_j)}} \tag{1}$$

where the similarity function $\text{sim}(\cdot,\cdot)$ is using dot product similarity to measure the similarity between the ith pixel in $\mathbf{M}$ and the jth pixel in $\mathbf{N}$. Meanwhile, feature $\mathbf{F}'$ is fed into another function λ to get a new feature $\mathbf{L} \in \mathbb{R}^{C\times H\times W}$ and reshape it to $\mathbf{L} \in \mathbb{R}^{C\times S}$. Then, matrix multiplication is performed between $\mathbf{L}$ and $\mathbf{W}'$ to get the detail-grained contextual information $\mathbf{H} \in \mathbb{R}^{C'\times S}$ of each pixel

$$h_i = \mu\left(\sum_{j=1}^{S} w'_j l_j\right) \tag{2}$$

where η, θ, λ, and μ are nonlinear transformation functions consisting of a 1×1 convolutional layer, a batch normalization layer, and an ReLU activation function.

B. P2O Subnetwork

The goal of the P2P subnetwork is to fill in the spatial details of objects. However, due to the diverse appearance of same-category objects (e.g., car), the features within the corresponding object regions often vary according to their positions. This demands us to capture more representative features for objects in each category. In order to achieve this goal, we design a P2O subnetwork. It introduces an object overall representation obtained by aggregating features at all positions of the same-category object regions. In comparison with the individual feature at each position, this one could offer a smooth and global representation for objects of a particular category in the image. Based on this overall representation, the P2O subnetwork learns P2O relation for classifying the objects. Note that an object is indeed not equivalent to a category here. The definition of the category is classes of all existing objects in the dataset, and object means the concrete items of the specific category. With respect to each input image, the number of existing categories is fixed, while the number of objects in a particular category varies. In the following, we elaborate on the process to generate the overall representation of objects in each category and model the P2O relation.

As illustrated in Fig. 2, we propose a clustering learning method to obtain the global descriptor of each category. More concretely, small patches (as shown in Fig. 3) with the spatial size of 32×32 are cropped from the original image to construct the training set, which ensures that almost all of them contain one dominant object. Next, each patch is fed into ResNet18 pretrained on the ImageNet dataset [27] to derive the corresponding features, and the principal component analysis (PCA) is utilized to reduce features to D dimensions for further clustering. Then, PCA-reduced feature vectors are sorted into K distinct groups with k-means based on the Euclidean distance measure. The cluster assignments are used as pseudolabels to retrain ResNet18. Ultimately, the clustering learning method iteratively groups the features and uses the subsequent assignments as supervision to update the weights of the network. The centroid derived from clusters is regarded as the global descriptor corresponding to each category. However, this type of alternating procedure is prone to encounter two kinds of poor cluster results, leading to the degradation of ResNet18 performance. Especially, all the inputs are likely assigned to a single cluster, or k-means assign the vast majority of images to a few clusters. Herein, we adopt two strategies to avoid such results. For the first case, we randomly select a nonempty cluster and use its centroid with a small random perturbation as the new centroid for the empty cluster. Then, the points belonging to the nonempty cluster are reassigned to the two resulting clusters. For the second case, we sample input images based on a uniform distribution over pseudolabels assigned by k-means.

Once obtained the global descriptors of all categories, they are demanded to be adapted to the specific scene (i.e., each sample). We assume that the global descriptor (i.e., centroid) of the kth category is denoted as $d_k \in \mathbb{R}^D$. Inspired by Zhou *et al.* [28], the category-specific attention maps can be obtained by fusing the global descriptor of each category and the convolutional feature via a multimodal fusion model. Specifically, given a convolutional feature $\mathbf{F} \in \mathbb{R}^{C'\times H\times W}$, we first project it and the kth category's global descriptor into

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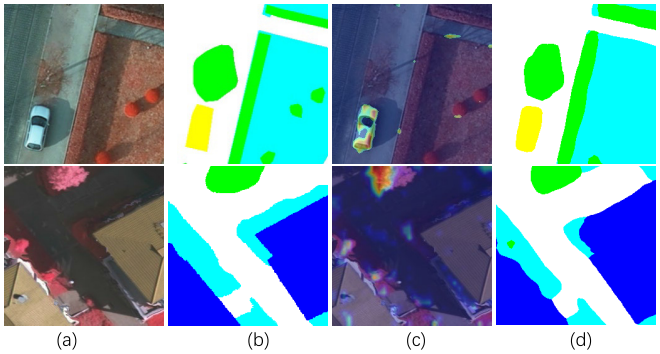


Fig. 4. Visualization results of category-specific (e.g., car and tree) regions. (a) Input image. (b) Ground truth. (c) Attention map. (d) Result.

a B-channel hidden feature $\mathbf{T}_k \in \mathbb{R}^{B\times H\times W}$

$$\mathbf{T}_k = \mathbf{Z}^\top \sigma(\mathbf{X}^\top \mathbf{F} \circ \mathbf{Y}^\top(\mathbf{1}d_k)) + b_1 \tag{3}$$

where $\mathbf{Z} \in \mathbb{R}^{E\times B}$, $\mathbf{X} \in \mathbb{R}^{C'\times E}$, and $\mathbf{Y} \in \mathbb{R}^{D\times E}$ are weight matrices, and $b_1 \in \mathbb{R}^B$ is a weight vector. E is the number of channels of the intermediate feature. "σ" and "$\circ$" represent the sigmoid function and the elementwise product operator, respectively. Note that $\mathbf{1}$ is a matrix of all ones, which is used to expand d_k into a $D \times H \times W$ tensor. The attention map can be inferred

$$\mathbf{A}_k = \text{Softmax}(w\mathbf{T}_k + b_2) \tag{4}$$

where $w \in \mathbb{R}^B$ and $b_2 \in \mathbb{R}$ prefer the learnable weights. $\mathbf{A}_k$ denotes the similarity between each pixel and the kth category. Some channels of category-specific maps are shown in Fig. 4. It can be observed that they could accurately locate the desired object regions of a specific category. The overall representation $r_k \in \mathbb{R}^{C'}$ of objects in the kth category can be derived by aggregating all the pixel features weighted by their similarities to the kth category

$$r_k = \sum_{i=1}^{H}\sum_{j=1}^{W} a_{k(i,j)} f_{(i,j)} \tag{5}$$

where $f_{(i,j)}$ is the feature of the position (i, j) in $\mathbf{F}$ and $a_{k(i,j)}$ is the similarity of the pixel located at the position (i, j) to the kth category.

To capture semantic-grained contextual information for each pixel, we compute the relation $\mathbf{W} \in \mathbb{R}^{S\times K}$ between each pixel and objects of K categories. We reshape $\mathbf{F} \in \mathbb{R}^{C'\times H\times W}$ to $\mathbf{F} \in \mathbb{R}^{C'\times S}$ and multiply it by the object overall representation r to obtain $\mathbf{W}$ as follows:

$$w_{ik} = \frac{e^{\text{sim}(f_i, r_k)}}{\sum_{j=1}^{K} e^{\text{sim}(f_i, r_j)}} \tag{6}$$

where $\text{sim}(f_i, r_k)$ is the similarity between the ith pixel and objects of the kth category. A common parameterization for the similarity function $\text{sim}(\cdot,\cdot)$ is using dot product similarity $\text{sim}(f_i, r_k) = \rho(f_i)^\top \delta(r_k)$, where ρ and δ are two nonlinear transformation functions. They all consist of a 1×1 convolutional layer, a batch normalization layer, and an ReLU activation function. This procedure is motivated by self-attention [29] for a better relation measurement. The semantic-grained contextual information $\mathbf{G} \in \mathbb{R}^{C'\times S}$ of each pixel is computed by aggregating K overall representations of all the categories according to its P2O relation

$$g_i = \phi\left(\sum_{k=1}^{K} w_{ik}\psi(r_k)\right). \tag{7}$$

Similar to ρ and δ, ϕ and ψ are nonlinear transformation functions composed of a 1×1 convolutional layer, a batch normalization layer, and an ReLU activation function.

In order to obtain the hierarchical contextual information, we reshape $\mathbf{G}$ and $\mathbf{H}$ to $\mathbf{G} \in \mathbb{R}^{C'\times H\times W}$ and $\mathbf{H} \in \mathbb{R}^{C'\times H\times W}$, and the elementwise sum is performed for feature aggregation. This hierarchical context is capable to offer precise category information and clear spatial details for objects. Finally, the final segmentation map is generated via a convolutional layer. The setting of hyperparameters is given in Table I.

TABLE I
HYPERPARAMETERS' SETUP OF THREE P2P AND P2O SUBNETWORKS ATTACHED TO THE CORRESPONDING LAYER OF RESNET18 ($V = 64$)

Layer	3	4	5
B	V	$2V$	$4V$
C	V	$2V$	$4V$
C'	$2V$	$4V$	$8V$
D	300	300	300
E	V	$2V$	$4V$

III. EXPERIMENTAL RESULTS

We conduct comprehensive experiments on the ISPRS Potsdam dataset [30], the ISPRS Vaihingen dataset [31], and GID to evaluate the proposed HCNet. In the following, we first introduce the datasets and implementation details, and then, we perform a series of ablation studies. Finally, we report our results on Potsdam, Vaihingen, and GID datasets.

A. Datasets

The Potsdam dataset consists of 38 high-resolution airborne images with a spatial dimension of 6000×6000 pixels. Each airborne image is composed of four channels (near-infrared, red, green, and blue). Similar to [32]–[36], 24 images are used for training, and the remaining 14 images are used for testing. The ground sampling distance of this dataset is 5 cm. The reference data are labeled with six land-cover types: impervious surfaces, building, low vegetation, tree, car, and background.

The Vaihingen dataset contains 33 airborne images collected over a 1.38-km^2 area of the city. Similar to [33], [35], [37], and [38], 16 images are used for training, and the remaining 17 images are used for testing. The ground sampling distance of this dataset is 9 cm. The average size of each image is 2494×2064 pixels, and each of them has three bands, corresponding to near-infrared, red, and green wavelengths. The reference data are divided into the same six categories as the Potsdam.

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The GID dataset contains 150 high-resolution Gaofen-2 images, covering a geographic area of 506 km^2. According to [26], 120 images are used for training, and the remaining 30 images are used for testing. The multispectral image used to establish GID provides a spectral range of blue, green, red, and near-infrared, and a spatial dimension of 6800×7200 pixels. Five major categories are annotated: built-up, farmland, forest, meadow, and water for pixelwise land-cover classification.

B. Implementation Details

Near-infrared, red, and green channels are used to constitute the input image for three datasets. To make full use of computing resources, a 256×256 sliding window is used to crop the input images for generating training samples during training. At the inference phase, the same sliding method is used to crop input tiles from the original testing image, and they are processed one by one. Two evaluation metrics [i.e., F1 score and overall accuracy (OA)] are utilized to evaluate the performance of methods. F1 score is defined as the harmonic mean of precision and recall. OA is the metric that considers the global aspects of classification.

For the clustering learning method, the stochastic gradient descent optimizer is applied. Specifically, the base learning rate, momentum, and weight decay are set to 0.01, 0.9, and 0.00001, respectively. We train it for 100 epochs. For the proposed HCNet, following [15], [21], and [23], we employ a poly learning rate policy [39] where the initial learning rate is multiplied by $(1 - (\text{iter}/\text{total_iter}))^{0.9}$ after each iteration. The base learning rate is set to 0.01 for three datasets. The momentum and weight decay coefficients are set to 0.9 and 0.0001, respectively. Besides, we set the training time to 200 epochs for each dataset. The loss of the proposed HCNet is set to cross-entropy. The network is implemented with PyTorch and trained on one NVIDIA TITAN X 12-GB GPU.

In the following experiments, we choose nine popular deep methods to validate the effectiveness of HCNet, including FCN-8s [10], UNet [11], SegNet [12], PSPNet [14], DeepLabv3+ [15], EncNet [23], DANet [21], CGFDN [28], and S-RA-FCN [24]. For PSPNet, the auxiliary loss weight α is selected from a given set $\{0, 0.3, 0.4, 0.6, 0.9\}$. For EncNet, the SE-loss weight α is chosen from $\{0, 0.1, 0.2, 0.4, 0.8\}$, and the number of code-words is selected from $\{0, 16, 32\}$. For CGFDN, decoupling loss weight λ is chosen from a given set $\{0, 0.001, 0.01, 0.1, 1, 10\}$.

C. Ablation Study

1) Effectiveness of Two Subnetworks: We employ two subnetworks to capture hierarchical contextual information for better scene segmentation. To verify their performance, we conduct experiments on the validation sets of ISPRS datasets with different settings in Table II. Specifically, P2O and P2P subnetworks are separately attached to ResNet18 rather than being used simultaneously. First, the summation operation of baseline FCN is replaced with the weighted summation, which brings 0.8% improvement on Potsdam and 0.5% improvement on Vaihingen. Then, compared with baseline FCN, employing P2O subnetwork yields a result of 96.7% and 97.8% in OA on two datasets, which separately brings 0.9% and 0.4% improvements. Meanwhile, employing the P2P subnetwork individually outperforms the baseline by 1.6% and 0.7%, respectively. When we integrate the two subnetworks together, the performance further improves to 98.3% and 98.6%, respectively. Experimental results demonstrate that the proposed two subnetworks make remarkable contributions to scene segmentation.

TABLE II
ABLATION STUDY (OA) OF HCNET ON THREE DATASETS

Model	Backbone	WS	P2O	P2P	Potsdam	Vaihingen
FCN	ResNet18	*w/o*			95.0	96.9
HCNet	ResNet18	*w*			95.8	97.4
HCNet	ResNet18	*w*	✓		96.7	97.8
HCNet	ResNet18	*w*		✓	97.4	98.1
HCNet	ResNet18	*w*	✓	✓	98.3	98.6

WS denotes weighted summation operation.

TABLE III
EFFICACY (OA) OF CATEGORY-SPECIFIC ATTENTION MAPS WITHIN THE P2P SUBNETWORK

Model	Potsdam	Vaihingen
FCN	95.8	97.4
FCN+P2P	97.0	97.9
FCN+P2P (s)	97.4	98.1

's' represents that the P2P sub-network utilizes category-specific attention maps.

TABLE IV
EFFICACY (OA) OF THE ADAPTATION OF CATEGORIES' GLOBAL DESCRIPTORS

Model	Potsdam	Vaihingen
HCNet (*w* adaptation)	98.3	98.6
HCNet (*w/o* adaptation)	97.7	98.2

Besides, we also discuss the effects of category-specific attention maps on the P2P subnetwork. As shown in Table III, we first attach a common P2P subnetwork to the baseline FCN without category-specific attention maps. As a result, it improves OA by 1.2% and 0.5% because it is capable to model the relation between pixels that are far away from each other. By augmenting the convolutional feature with category-specific attention maps, the difference among pixels is enhanced, thus improving OA by 0.4% and 0.2%. Furthermore, to verify the efficacy of the adaptation of the global descriptors, we conduct the experiments on three datasets with two kinds of HCNets. Without adaptation, it can be observed from Table IV that the performance of HCNet decreases by 0.6% and 0.4%. We conjecture that the overall representation of objects in each category varies according to different scenes (i.e., input samples), which means that it is necessary to adapt the category's global descriptor to the specific scene.

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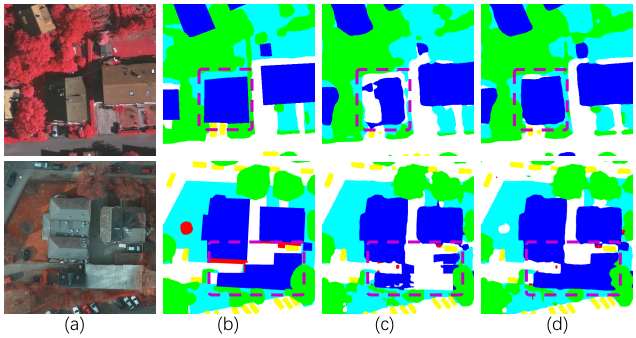


Fig. 5. Visualization results of the P2O subnetwork. (a) Input image. (b) Ground truth. (c) Without P2O. (d) With P2O.

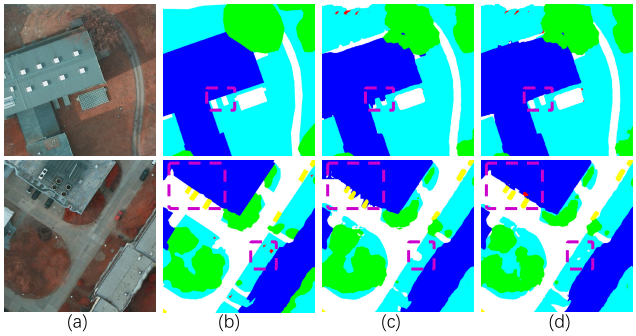


Fig. 6. Visualization results of the P2P subnetwork. (a) Input image. (b) Ground truth. (c) Without P2P. (d) With P2P.

2) Visualization: The effects of the P2O subnetwork can be visualized in Fig. 5. Without this subnetwork, a large portion of the *building* is wrongly classified. When integrating semantic-grained contextual information, those misclassified regions are corrected, improving intraobject semantic consistency. With respect to the P2P subnetwork, Fig. 6 demonstrates that it can help obtain clearer details and sharp boundaries, such as the building in the first and second rows and the low vegetation in the second and third rows.

3) Comparison With Related Deep Methods: To evaluate the effectiveness of the proposed HCNet, comparisons with two related deep methods, including DANet [21] and CGFDN [28], are presented. Statistics in Tables V and VI show the segmentation accuracy of HCNet, DANet, and CGFDN on the testing sets of Potsdam and Vaihingen. As P2P relation-based methods, DANet and CGFDN achieve worse performance than HCNet. Concretely, compared with the second best model CGFDN, HCNet obtains increments of 0.5% and 0.4% in OA on Potsdam and Vaihingen. With the aggregation of semantic-grained contextual information, HCNet improves the intraobject semantic consistency. Meanwhile, HCNet captures detail-grained contextual information for better preservation of the spatial details. Benefiting from this, HCNet enhances the OA of pixelwise semantic segmentation and the F1 metrics on easily confused categories (e.g., *impervious surface* and *building*). Some segmentation results are illustrated in Fig. 7. DANet and CGFDN fail to correctly classify the marked *building*, whereas our method makes relatively accurate predictions. We conjecture that the appearance of *building* is visually similar to neighboring *impervious surface* in this scene, leading to the misclassification of comparison methods. Due to the hierarchical context, HCNet improves the intraobject semantic consistency and obtains clear and sharp boundaries.

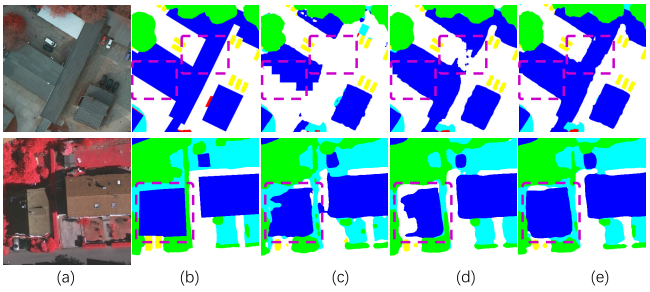


Fig. 7. Qualitative comparisons between HCNet and related deep methods on ISPRS datasets. (a) Input Image. (b) Ground truth. (c) DANet. (d) CGFDN. (e) HCNet.

TABLE V
QUANTITATIVE COMPARISONS BETWEEN HCNET AND RELATED DEEP METHODS ON POTSDAM

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1
DANet [21]	91.6	95.4	85.1	86.8	94.2	89.7	90.6
CGFDN [28]	92.1	95.6	86.3	87.9	94.9	90.3	91.4
HCNet	92.5	96.2	86.7	88.3	95.1	90.8	91.8

TABLE VI
QUANTITATIVE COMPARISONS BETWEEN HCNET AND RELATED DEEP METHODS ON VAIHINGEN

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1
DANet [21]	91.0	94.5	80.9	88.1	83.9	90.0	87.7
CGFDN [28]	91.9	95.0	81.5	88.7	85.0	90.6	88.4
HCNet	92.2	95.5	82.2	88.9	85.9	91.0	88.9

TABLE VII
QUANTITATIVE COMPARISONS (OA) WITH DIFFERENT V VALUES

V	16	32	64	128	256
Potsdam	97.1	97.6	98.3	98.0	98.1
Vaihingen	97.9	98.2	98.6	98.4	98.4

TABLE VIII
QUANTITATIVE COMPARISONS (CALINSKI–HARABAZ SCORE) WITH K-MEANS BASED ON FOUR DISTANCES

Distance	Euclidean	Manhattan	Chebyshev	Cosine
Potsdam	5694	2840	3226	2542
Vaihingen	6298	4084	4798	3392

D. Hyperparameter Selection

There are three important hyperparameters in HCNet, including V, distance measure, and the size of input patches

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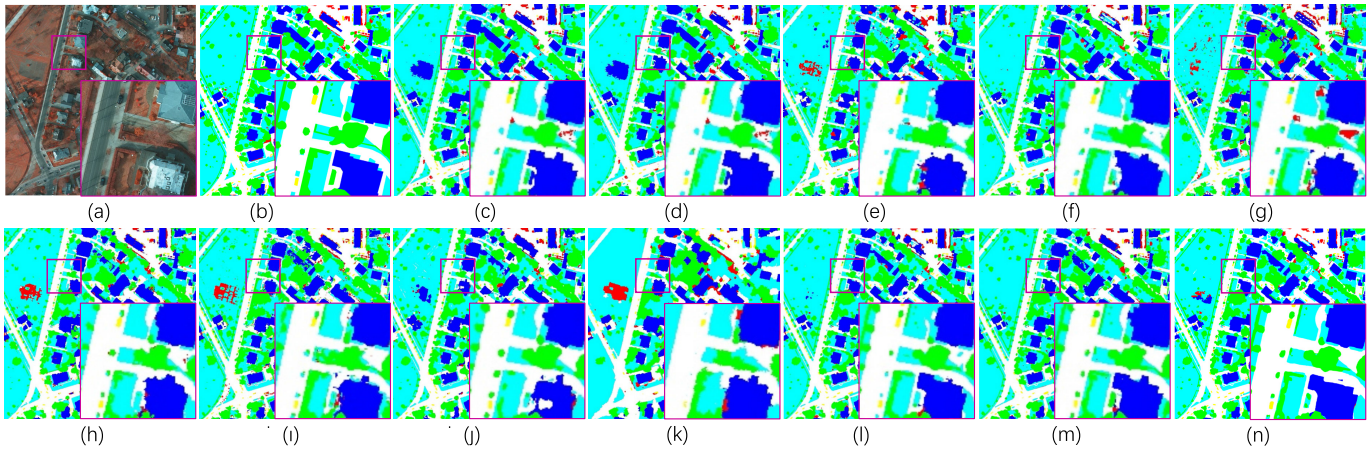


Fig. 8. Segmentation results produced by benchmark models on a typical image from the Potsdam dataset. Legend: white: impervious surfaces; blue: building; cyan: low vegetation; green: tree; yellow: car; and red: background. (a) Image. (b) Ground truth. (c) BUCTY5. (d) CAS_Y3. (e) KLab_2. (f) CVEO2. (g) UFMG_4. (h) RIT_L7. (i) UZ_1. (j) DST_5. (k) SVL_1. (l) CASIA2. (m) SWJ_2. (n) HCNet.

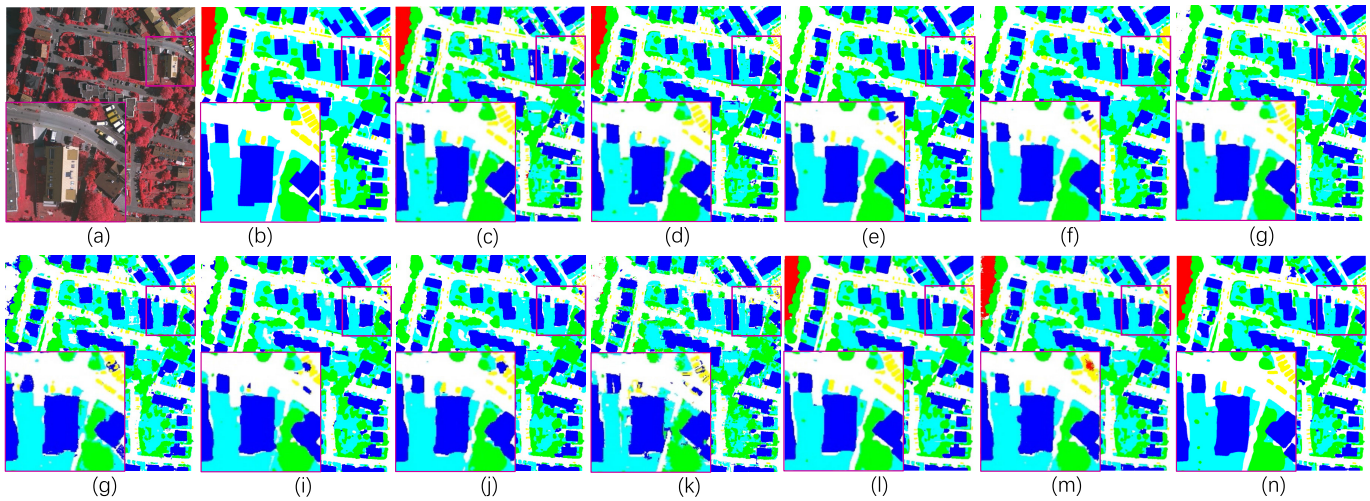


Fig. 9. Segmentation results produced by benchmark models on a typical image from the Vaihingen dataset. Legend: white: impervious surfaces; blue: building; cyan: low vegetation; green: tree; yellow: car; and red: background. (a) Image. (b) Ground truth. (c) CVEO3. (d) UFMG_4. (e) DLR_10. (f) ADL_3. (g) ONE_7. (h) RIT_L8. (i) UZ_1. (j) DST_2. (k) SVL_3. (l) CASIA2. (m) HUSTW5. (n) HCNet.

TABLE IX
QUANTITATIVE COMPARISONS (OA) WITH DIFFERENT INPUT SIZES FOR CLUSTERING LEARNING

Size	8×8	16×16	32×32	64×64	128×128
Potsdam	96.8	97.6	98.3	97.8	97.0
Vaihingen	97.3	98.1	98.6	97.7	96.1

for clustering learning. Similar to [43], we construct the validation sets of ISPRS Potsdam and Vaihingen by, respectively, choosing 14 and 8 images from the corresponding training sets, and the hyperparameter selection is performed on these two validation sets. In the hyperparameter selection, the OA and the Calinski–Harabaz score are selected to evaluate the performance of different hyperparameters. For the first factor, we fix other factors to find the optimal V from a candidate set $\{16, 32, 64, 128, 256\}$. Table VII demonstrates the effects of different V values on OA of Potsdam and Vaihingen datasets. As shown in this table, OA will first improve and then tends to be stable, as V increases. HCNet obtains the best performance when $V = 64$; thus, we choose 64 as the optimal V. Second, to find out what kind of distance is suitable for clustering learning, we perform k-means with four different distances on the validation sets of ISPRS datasets. Herein, we choose four common distances, including the Euclidean distance, the Manhattan distance, the Chebyshev distance, and the cosine distance. Experimental results of four kinds of k-means are listed in Table VIII. To quantitatively evaluate the clustering performance, we calculate the Calinski–Harabaz score $s = ((\mathrm{tr}(\mathbf{S}_B))/(\mathrm{tr}(\mathbf{S}_W)))((N-K)/(K-1))$, where $\mathbf{S}_B$ and $\mathbf{S}_W$, respectively, denote interclass divergence matrix and intraclass divergence matrix, N indicates the number of input patches to be clustered, and K is the number of clusters. The higher the value of s, the larger the clustering quality. From this table, we can observe that the Euclidean distance helps k-means achieve the best s. Finally, we fix other

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TABLE X
QUANTITATIVE COMPARISONS WITH BENCHMARK MODELS ON THE POTSDAM DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
BUCTY5 [36]	93.1	97.3	86.8	87.1	94.1	90.6	91.7	-
CAS_Y3 [34]	92.2	95.7	87.2	87.6	95.6	90.1	91.7	-
KLab_2 [32]	89.7	92.7	83.7	84.0	92.1	86.7	88.4	-
CVEO2 [33]	91.3	94.5	86.5	87.7	95.1	89.0	91.0	-
UFMG_4 [35]	90.8	95.6	84.4	84.3	92.4	87.9	89.5	-
RIT_L7 [37]	91.2	94.6	85.1	85.1	92.8	88.4	89.8	-
UZ_1 [40]	89.3	95.4	81.8	80.5	86.5	85.8	86.7	-
DST_5 [41]	92.5	96.4	86.7	88.0	94.7	90.3	91.7	-
SVL_1 [42]	83.5	91.7	72.2	63.2	62.2	77.8	74.6	-
CASIA2 [43]	93.3	97.0	87.7	88.4	96.2	91.1	92.5	-
SWJ_2 [44]	94.4	97.4	87.8	87.6	94.7	91.7	92.4	-
FCN-8s [10]	89.9	93.7	83.0	85.2	93.5	87.8	89.1	71.7
UNet [11]	88.2	91.1	82.8	84.9	91.6	86.2	87.7	68.5
SegNet [12]	87.8	90.7	81.0	84.7	89.7	85.7	86.8	66.2
PSPNet [14]	91.0	94.8	84.7	86.3	90.8	88.9	89.5	72.3
DeepLabv3+ [15]	91.3	94.8	84.2	86.6	93.8	89.2	90.1	73.8
EncNet [23]	91.0	94.9	84.4	85.9	93.6	89.0	90.0	73.2
S-RA-FCN [24]	90.7	94.2	83.8	85.8	93.6	88.5	89.6	72.5
HCNet	92.5	96.2	86.7	88.3	95.1	90.8	91.8	76.4

TABLE XI
QUANTITATIVE COMPARISONS WITH BENCHMARK MODELS ON THE VAIHINGEN DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
CVEO3 [33]	91.0	93.0	81.3	88.3	83.0	88.6	87.3	-
UFMG_4 [35]	91.1	94.5	82.9	88.8	81.3	89.4	87.7	-
DLR_10 [38]	92.3	95.2	84.1	90.0	79.3	90.3	88.2	-
ADL_3 [45]	89.5	93.2	82.3	88.2	63.3	88.0	83.3	-
ONE_7 [46]	91.0	94.5	84.4	89.9	77.8	89.8	87.5	-
RIT_L8 [37]	89.6	92.2	81.6	88.6	76.0	87.8	85.6	-
UZ_1 [40]	89.2	92.5	81.6	86.9	57.3	87.3	81.5	-
DST_2 [41]	90.5	93.7	83.4	89.2	72.6	89.1	85.9	-
SVL_3 [42]	86.6	91.0	77.0	85.0	55.6	84.8	79.0	-
CASIA2 [43]	93.2	96.0	84.7	89.9	86.7	91.1	90.1	-
HUSTW5 [47]	93.3	96.1	86.4	90.8	74.6	91.6	88.2	-
FCN-8s [10]	90.0	93.0	77.7	86.5	80.4	88.3	85.5	75.5
UNet [11]	90.5	93.3	79.6	87.5	76.4	89.2	85.5	75.5
SegNet [12]	90.2	93.7	78.5	85.8	83.9	88.5	86.4	76.8
PSPNet [14]	90.8	94.8	80.7	88.2	73.0	89.9	85.5	75.7
DeepLabv3+ [15]	91.4	94.7	79.6	87.6	85.8	89.9	87.8	79.0
EncNet [23]	91.2	94.1	79.2	86.9	83.7	89.4	87.0	77.8
S-RA-FCN [24]	90.5	93.8	79.6	87.5	82.6	89.2	86.8	77.3
HCNet	92.2	95.5	82.2	88.9	85.9	91.0	88.9	80.7

hyperparameters and select the optimal size of input patches from a set $\{8 \times 8, 16 \times 16, 32 \times 32, 64 \times 64, 128 \times 128\}$. As shown in Table IX, HCNet achieves the highest OA on Potsdam and Vaihingen datasets when the size is set to 32×32, while OA decreases as the size becomes larger or smaller. We conjecture that, when the size is big, the pixels in the input patch are likely from two or more categories. As a result, it is difficult to learn discrimination between different categories. In addition, if the size is too small, the feature map downsampled by multiple convolutional/max-pooling layers loses abundant spatial information. Thus, when the size is set to 16×16 and 8×8, the performance of HCNet also degrades.

E. Results on ISPRS Datasets

To better evaluate the performance of HCNet on ISPRS datasets, we compare it with not only 15 state-of-the-art benchmark models but also seven popular deep methods. For benchmark models, we select DLR_10 [38], ONE_7 [46], ADL_3 [45], BUCTY5 [36], CAS_Y3 [34], KLab_2 [32], CVEO* [33], UFMG_4 [35], RIT_L* [37], DST_* [41], UZ_1 [40], SVL_* [42], CASIA2 [43], SWJ_2 [44], and HUSTW5 [47] as the comparison methods. The experimental results are listed in Tables X and XI. It can be observed from Table X that HCNet outperforms most of the competing methods due to its exploration of hierarchical contextual information. However, the performance of HCNet is inferior to those of CASIA2 and SWJ_2. We conjecture that both of them utilize a more complicated backbone (i.e., ResNet101) and propose inference strategies, such as multiscale tests, thus achieving a better performance. Besides, SWJ_2 further improves OA by 0.6% than CASIA2 due to a synergistic use of multilevel and multiscale deep features. As shown in Fig. 8, most comparison models produce many fragmented segments and obtain blurred boundaries of objects. For instance, a portion of the building is misclassified into the road, and the

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TABLE XII
PERFORMANCE ACHIEVED BY DIFFERENT DEEP METHODS ON THE VALIDATION SET OF THE POTSDAM DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
FCN-8s [10]	95.1	98.1	91.8	90.0	87.9	95.0	92.6	84.2
UNet [11]	92.6	97.5	86.0	85.2	85.1	92.2	89.3	75.5
SegNet [12]	88.9	93.9	80.6	81.8	82.4	88.2	85.5	68.4
PSPNet [14]	97.4	99.1	95.4	93.9	87.3	97.1	94.6	88.2
DeepLabv3+ [15]	92.3	95.5	89.3	90.4	87.8	92.2	91.1	78.9
EncNet [23]	96.0	97.9	93.8	92.8	90.1	94.8	94.1	84.0
S-RA-FCN [24]	95.9	97.2	94.4	93.1	88.0	95.6	93.7	85.5
DANet [21]	96.8	98.9	93.1	93.9	91.1	93.8	94.8	83.8
CGFDN [28]	96.8	98.9	93.7	91.2	88.4	96.2	93.8	87.4
HCNet	98.5	99.5	97.1	95.8	90.5	98.3	96.3	93.2

TABLE XIII
PERFORMANCE ACHIEVED BY DIFFERENT DEEP METHODS ON THE VALIDATION SET OF THE VAIHINGEN DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
FCN-8s [10]	97.1	98.6	92.6	97.2	95.7	96.9	96.2	93.0
UNet [11]	95.3	98.3	90.5	93.5	83.1	95.3	92.1	86.1
SegNet [12]	94.7	95.7	90.7	95.7	96.1	94.3	94.6	90.0
PSPNet [14]	97.3	99.0	94.8	96.5	84.4	97.2	94.4	89.9
DeepLabv3+ [15]	97.3	98.0	90.3	93.2	96.6	95.6	95.1	91.0
EncNet [23]	95.3	96.7	89.7	93.2	93.1	94.1	93.6	88.2
S-RA-FCN [24]	93.9	95.8	91.2	94.1	91.3	94.2	93.3	87.5
DANet [21]	96.4	97.0	93.1	97.0	94.7	96.3	95.6	91.8
CGFDN [28]	94.5	97.6	85.8	90.9	90.7	93.7	91.9	83.5
HCNet	98.8	99.6	97.3	97.9	97.0	98.6	98.1	96.3

boundaries between the car and the road are ambiguous. After introducing hierarchical contextual information, our model successfully distinguishes the building and the road, and obtains sharp boundaries. Nevertheless, HCNet does not focus on making great use of multiscale context, obtaining some commission errors, especially for tiny patches of low vegetation close to the building. Similar results are demonstrated in Table XI and Fig. 9 on the testing set of Vaihingen. Again, HCNet, CASIA2, and HUSTW5 perform better than other competing methods. Based on more complicated backbones and inference strategies, HUSTW5 and CASIA2 obtain higher OA compared with HCNet. For deep methods, FCN-8s [10], UNet [11], SegNet [12], PSPNet [14], DeepLabv3+ [15], EncNet [23], and S-RA-FCN [24] are chosen as the comparison methods. Tables XII and XIII list experimental results of all methods on the validation sets of Potsdam and Vaihingen datasets. It shows that the proposed HCNet achieves the best performance in terms of OA, mean F1, and mean IoU among all the comparison methods. Table X reports the numerical results achieved on the testing set of Potsdam. It can be observed that HCNet outperforms other methods with respect to mean F1 and OA. Similarly, Table X shows that HCNet achieves the best performance.

In addition, based on ISPRS datasets, we also analyze the time and space complexity of the proposed HCNet. Specifically, HCNet costs 29 ms to infer an image patch (256 × 256 pixels) with a GTX TITAN X GPU on Ubuntu. As shown in Fig. 10(a), HCNet (marked with a red triangle)

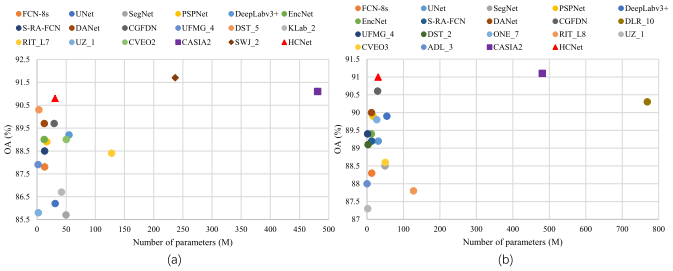


Fig. 10. Number of parameters and segmentation accuracy of HCNet and comparison methods. (a) Potsdam. (b) Vaihingen.

outperforms all deep methods within the acceptable range of the number of parameters on the Potsdam dataset. Besides, compared with the best method SWJ_2 (marked with brown rhombus) and the second best method CASIA2 (marked with a purple square), although HCNet achieves a little lower segmentation accuracy (0.9% ↓ and 0.3% ↓), it tremendously reduces the number of parameters by almost eight times and 15 times, respectively. The same conclusion can be drawn from Fig. 10(b). HCNet obtains a relatively good performance at the cost of tiny complexity on the Vaihingen dataset.

F. Results on GID

We carry out experiments on GID to further evaluate the effectiveness of our model. The comparison methods include

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TABLE XIV
PERFORMANCE ACHIEVED BY DIFFERENT DEEP METHODS ON THE GID DATASET

Method	Built-up	Farmland	Forest	Meadow	Car	OA	Mean F1	Mean IoU
FCN-8s [10]	82.7	92.5	79.9	3.9	90.3	95.6	69.9	64.8
UNet [11]	89.2	94.2	83.4	20.1	90.7	96.5	75.5	70.2
SegNet [12]	88.6	93.4	83.2	19.3	90.8	96.4	75.1	69.7
PSPNet [14]	87.8	95.5	88.2	40.1	92.4	97.5	80.8	76.0
DeepLabv3+ [15]	91.0	95.9	87.9	48.6	92.7	97.6	83.2	78.2
EncNet [23]	90.1	94.6	88.0	46.4	93.4	97.3	82.5	77.5
S-RA-FCN [24]	90.8	96.6	89.2	44.0	94.0	97.9	82.9	78.7
DANet [21]	91.8	96.8	90.4	47.3	96.0	98.0	84.5	80.4
CGFDN [28]	92.3	95.8	90.4	50.2	94.4	98.0	84.6	80.6
HCNet	92.0	97.1	88.3	51.7	93.9	98.2	84.6	80.5

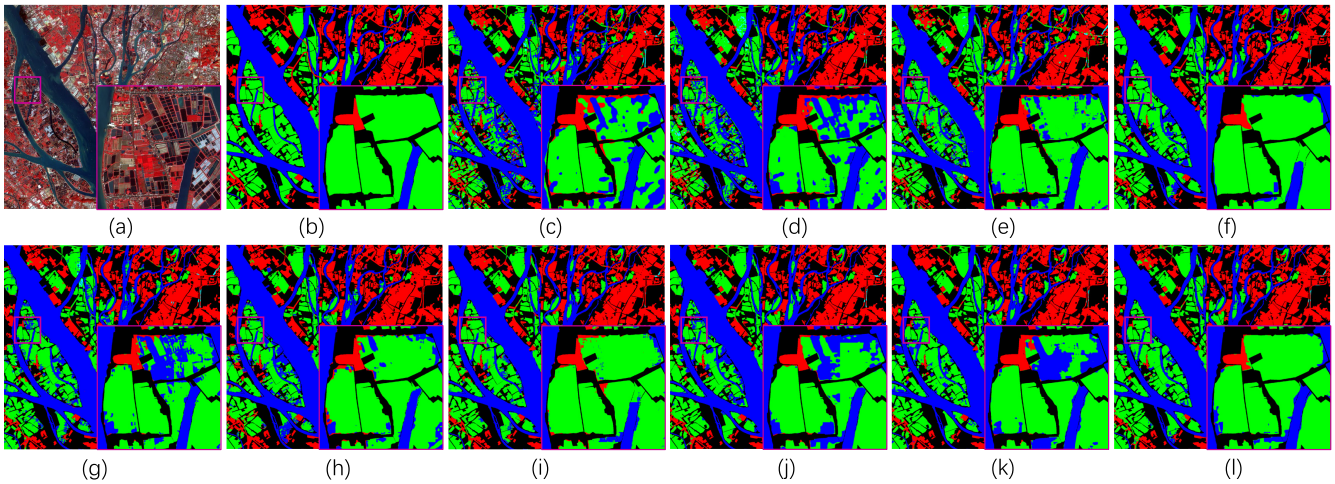


Fig. 11. Segmentation results produced by deep methods on a typical image from the GID dataset. Legend: red: built-up; green: farmland; cyan: forest; yellow: meadow; blue: water; and black: background. (a) Image. (b) Ground truth. (c) FCN-8s. (d) UNet. (e) SegNet. (f) PSPNet. (g) DeepLabv3+. (h) EncNet. (i) S-RA-FCN. (j) DANet. (k) CGFDN. (l) HCNet.

SegNet [12], UNet [11], FCN-8s [10], DeepLabv3+ [15], PSPNet [14], S-RA-FCN [24], EncNet [23], CGFDN [28], and DANet [21]. Table XIV exhibits quantitative results of GID. The baseline (FCN-8s) yields OA 95.6%. HCNet boosts the performance to 98.2%. Among comparison works, DANet learns the spatial- and channel-dimension semantic interdependencies, and CGFDN utilizes the co-occurrence relation to enhance the P2P relation, achieving the best performance. Different from comparison methods, we introduce P2P and P2O subnetworks to capture detail-grained and semantic-grained contextual information definitely, leading to the better performance of HCNet. Visual comparison examples are shown in Fig. 11. It can be seen that *farmland* is the most difficult class to be recognized in this image. *Water* and *farmland* categories are seriously confused by the comparison methods. This is because the visual appearances of these classes are similar. In comparison with other methods, the proposed HCNet successfully distinguishes these two categories due to its remarkable capacity of exploiting hierarchical context (including semantic and detailed information of objects) to solve visual ambiguities.

IV. CONCLUSION

In this article, two effective subnetworks are introduced to explore the hierarchical context for airborne image segmentation. Specifically, the P2O subnetwork models the P2O relation to offer valuable semantic information of the object. Meanwhile, the P2P subnetwork learns the P2P relation to preserve spatial details (e.g., boundary) of objects. Extensive ablation studies show that hierarchical context could help to produce more coherent and precise segmentation results. Comparison experiments on Potsdam and Vaihingen have demonstrated that the proposed HCNet represents a good balance between complexity and performance. Furthermore, HCNet shows a competitive segmentation result on another challenging dataset GID.

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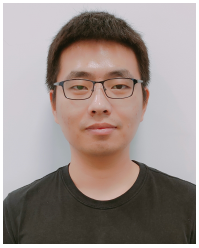
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